

Course: EGR 3350, Section 07

Technical Communications for Engineers and Computer Scientists

Instructor: Ms. Alysoun Taylor-Hall

Assignment: RFP

Title: Floating Solar Farms in the Ocean

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Abstract

This proposal will present a new design to improve the efficiency and lifespan of floating solar farms in ocean environments. Floating solar farms offer many advantages to land based solar farms which include: reducing land usage, lower installation costs, and increased energy efficiency. However, there are challenges in the ocean environment which cause problems as to how well offshore farms work. The biggest challenges are corrosion, dirty panel surfaces, and the harsh conditions of the ocean. All these challenges significantly reduce the system's performance and durability.

To combat these challenges, this project proposes a design prototype that integrates three main components: specialized materials, robust mooring system, and an automated monitoring and cleaning system. The design includes high-density polyethylene (HDPE) with an anti-corrosive epoxy glass coating to protect the panels from the saltwater. A flexible mooring system with anchors, rubber chains and wave barriers will keep the solar panel platform stable while simultaneously reducing structural damage from the motion of the water. The biggest innovation for this proposal is the IoT-based monitoring and cleaning system. Sensors will constantly track the solar panels' performance to find any reduced efficiency due to dirt, salt, algae and any other debris. When a panel's performance drops below a given threshold, the detection system will send the panel cleaning robot to the affected area. This system reduces the need for manual maintenance, optimizes the solar energy collected, and extends the lifespan of the panels.

Key Terms: Floating solar farm, Solar panel efficiency, Automated cleaning system, Energy optimization, Offshore solar power, Maintenance reduction

Introduction

Purpose

The purpose of this proposal is to improve the efficiency of floating solar farms by designing a sustainable prototype.

Background

There has been a strong interest in developing and using cleaner energy sources. One common method of creating cleaner energy is by converting sunlight into electrical energy using solar panels. Solar panels are typically installed on land. This, however, takes up available land space. This is why floating solar farms are being developed by installing solar panels on bodies of water. According to Nedelcu et al. (2026), solar panels on water ultimately save land for things such as agriculture, housing, and infrastructure. Installing solar panels on water also reduces cost due to the installation being cheaper than purchasing land and installing solar panels (Nedelcu et al., 2026).

Floating solar farms also offer a better performance compared to land-based systems. According to Huang et al. (2025), floating solar panels are approximately 5% more efficient compared to land-based systems due to the surrounding water cooling the solar panels down. Floating solar farms are also able to absorb more sunlight than land-based systems due to the sunlight reflecting off of the surrounding water (Huang et al., 2025).

The challenge of designing floating solar panels is that they must be able to withstand environmental conditions, especially ocean conditions. While floating solar farms already have success in calm waters, the installation and maintenance become challenging when placing them in ocean environments. Factors such as saltwater causing corrosion, debris accumulation, strong winds, and strong waves can damage the system and reduce the system's efficiency (Huang et al., 2025; Thangarasu et al., 2024).

Scope

This proposal addresses the design of floating solar panels and how materials, the mooring system, and a cleaning system which will improve the system's overall efficiency and viability in the ocean. A prototype cleaning robot will be created.

Discussion

Approach

The cost of offshore solar farms is influenced by several key factors, including the need to construct durable floating platforms, select corrosion-resistant materials, and implement reliable anchoring systems capable of withstanding ocean conditions. In addition, power transmission infrastructure must be developed to efficiently deliver energy back to shore. Within this project, the most significant cost will come from the development of the automated solar panel cleaning system. Because this component is new and requires extensive research, testing, and refinement, it represents a major portion of the initial investment. However, this investment is essential, as maintaining clean panels is critical for maximizing efficiency and ensuring consistent energy output over time.

Just as significant to costs is site selection. The strategic selection of a location is vital to ensuring optimal performance and feasibility. The most attractive sites for offshore solar power generation include locations where there are high levels of solar radiation, relatively calm bodies of water, and proximity to energy infrastructure facilities. Nearshore and coastal locations provide the perfect balance by cutting down on transmission expenses yet still ensuring consistent operation. Furthermore, this type of site selection also avoids potential problems

associated with land use issues, which makes offshore solar power generation highly beneficial in areas that are highly urbanized or densely populated.

The monitoring and cleaning system will improve floating solar farm efficiency by detecting when panels become dirty and triggering a cleaning response. Solar panels in ocean environments are exposed to salt water, algae, and debris, which accumulate on panel surfaces and reduce energy output by 10-25% (Mani & Pillai, 2010). To address this, an IoT-based software monitoring system will be installed on the floating platform to continuously track energy output from each panel. When the system detects a significant drop in efficiency, it will send an alert to operators indicating that cleaning is required.

The monitoring system will use sensors to collect real-time data including voltage, current, and power output from the solar panels (Kekre & Gawre, 2017). This data will be transmitted remotely, allowing operators to monitor farm performance without physical access to the offshore platform. When efficiency drops below a set threshold, the system will trigger an automated cleaning mechanism or notify maintenance teams. This approach reduces manual labor, prevents long-term efficiency loss, and extends the operational lifespan of the floating solar farm.

There needs to be a cleaning system for when the panels get dirty as this can significantly decrease their performance. The cleaning needs to be gentle to preserve the life span of the panels as aggressive or excessive cleaning could cause damage to the panels.

We propose to build a prototype robot with the following features: water jets, spinning brushes, wheels or tracks, and a control interface. The water jets will be fed by a pump using fresh water and will spray the dirty panels that the detection system determined. The robot will then use spinning circular brushes to gently clear the debris from the effected panels. We will do

the necessary research to determine the best way for the robot to move around using either wheels or tracks. A motor will be necessary to drive either system.

The materials used to design the floating solar farm are important to increase the system's overall viability over time. According to Xiong et al. (2025), the selected materials need to withstand the harsh ocean conditions such as saltwater exposure, UV rays, strong winds, and strong waves. These conditions are important to consider because they reduce the system's lifespan and durability, while increasing the risk of corrosion (Xiong et al., 2025). To address these conditions, the proposed design will consist of high-density polyethylene (HDPE) and anti-corrosive modified epoxy glass coating.

According to Nedelcu et al. (2026), HDPE is the most common material used for inshore floating solar farms. HDPE is a durable and buoyant material that can withstand UV rays and variable temperatures. However, this material should be combined with an anti-corrosive coating when designing an offshore system (Nedelcu et al. 2026). The anti-corrosive coating that will be used in this design is a modified epoxy glass coating. According to Xiong et al. (2025), the modified epoxy glass coating improves the lifespan to 18 months, nearly double of most anti-corrosive coatings. It also has an impedance modulus of $1e6 \Omega\text{cm}^2$, which indicates strong resistance to corrosion (Xiong et al., 2025). This coating can be reapplied every 18 months to continually protect the farm.

Floating solar farms need to be kept in place and above water after they are set up. It is important that the panels cannot drift very far from where they are supposed to be and don't get covered by water if the water level rises. If the floating panels have more freedom to move this can lead to micro-cracks in the modules which causes loss in durability, less buoyancy, and

endangers the safety of the solar system (Zeng et al., 2023). To avoid these problems a well-developed mooring system is needed to prevent any unnecessary damage to the panels.

We propose a mooring system that includes an anchor, rubber chains, and wave barriers. The anchor will be placed on the sea floor and allow for the chains to be attached to something. Having the mooring chains be made of rubber allows for better tension and are not as taught as steel. These chains need to have some freedom to allow for small movements in the platforms the panels are sitting on. A wave barrier would be placed just below the platforms. The barriers would dilute the power of incoming waves. This protects the platforms and will help to extend their lives.

Result

At the end of the contract term, this project will deliver an improved design for floating solar farms. This design will include corrosion-resistant materials, a mooring system capable of withstanding harsh ocean conditions, a detection system that determines when the solar panels need to be cleaned, and a cleaning system that automatically cleans the solar panels. The final outcome of this project will produce more sustainable energy by increasing the overall efficiency, durability, and lifespan of floating solar farms. These improvements benefit the Road Runner Research Foundation by reducing maintenance costs, increasing system durability, and supporting the public by producing cleaner energy.

Statement of Work

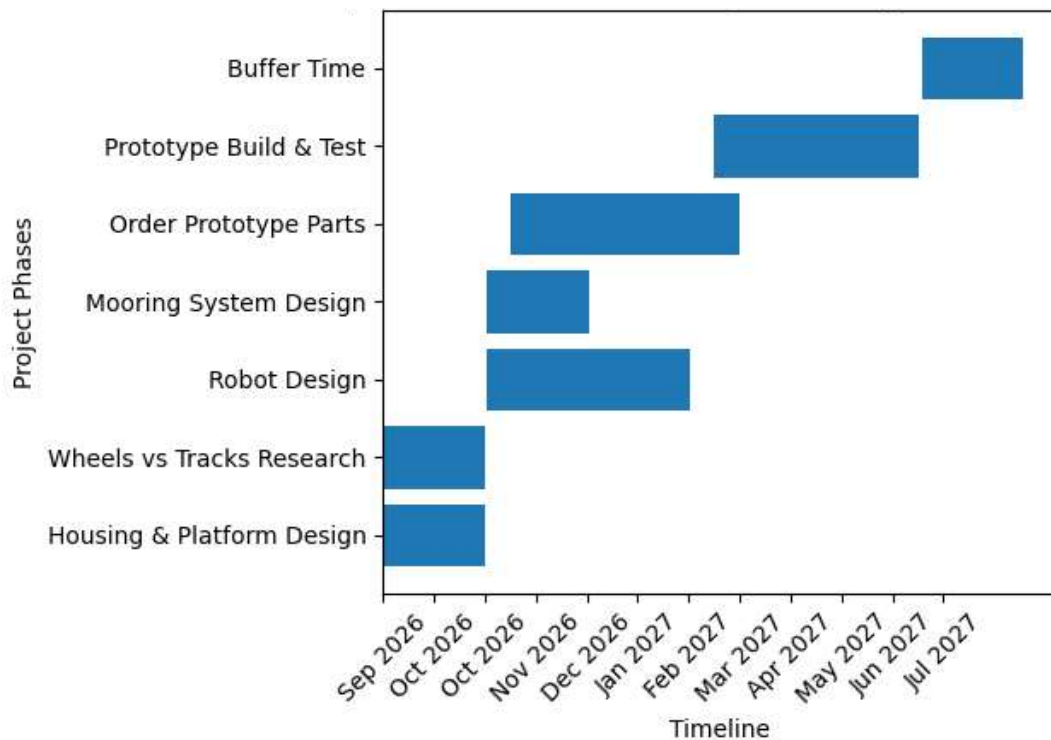
To achieve the goals of this proposal, the following tasks will be accomplished:

- **Task 1:** Design floating platform for the solar panels, material selection and structural layout for offshore location. (*2 months*)

- **Task 2:** Conduct research to determine the movement system of the cleaning robot, choosing either wheels or track systems. *(2 months, concurrent with Task 1)*
- **Task 3:** Design the automated solar panel cleaning robot. *(4 months)*
- **Task 4:** Design the mooring system, including anchors, rubber lines, and wave barrier. *(2 months, concurrent with Task 3)*
- **Task 5:** gather materials and components required for the robot prototype. *(concurrent with Task 3)*
- **Task 6:** Build and test the prototype, including performance evaluation. *(4 months)*
- **Task 7:** Perform final evaluation and adjust system to ensure reliability and completion of the project. *(2 months)*

Timeline

First, we will start designing the housing and platform that will contain the solar panels in September of 2026. This will take a maximum of 2 months to complete. During this same time research will be done on whether the cleaning robot should use wheels or tracks. Next, we will start designing the cleaning robot based on whether wheels or a track system was chosen. This process will take a maximum of 4 months. At the same time we will start designing the mooring system for the solar farm. This process will take a maximum of 2 months to complete. While the cleaning robot is being designed, we will start to order parts for the prototype so that we can start working on it when the design is finished. When the design is finished for the cleaning robot we will start building and testing the prototype. This will take a maximum of 4 months to complete. We will have an excess of two months just in case the timeline doesn't go as predicted.



(OpenAI, 2026)

Conclusion

Summary

The proposed designs will make solar farms more viable and cost effective for ocean conditions. Our research will deliver: A design for solar farm housing using ocean resistant materials; A design for a coastal based mooring system that will keep the panels secure; A design for a detection system that will sense which panels need to be cleaned; A prototype cleaning robot. As land space becomes less available, these design innovations will help solar panels in the ocean output enough power to become a viable option for renewable energy.

Utsav Acharya

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I am a Computer Science student with a strong academic record and hands-on project experience. I am a fast learner who enjoys and thrives in fast-paced environments. I'm dedicated to continued improvement, personally and professionally.

Education

Wright State University, Dayton, OH **May 2024-present**

Bachelor of Science in Computer Science

Coursework for this degree, which is focused on software development, systems, and mathematics which includes Computer Organization (Assembly), Discrete Math, Calculus I & II, Operating System, Computer Science I & II (Java), Statistics for Computer Scientists, Elementary Linear Algebra, Logic for Comp Scientist, Intro to Cyber Security, Data Struc & Algorithms, Tech Comm for EGR & CS.

Experience

Wright State University, Math Learning Center

The Math Learning Center is part of Wright State University's Student Success Center, offering free peer tutoring services to support students across a range of mathematics courses. In my current role, I provide one-on-one and group tutoring to help students succeed in their coursework, apply CRLA Level I certified instructional techniques to improve learning comprehension, and adapt my tutoring style to diverse student learning needs while maintaining an encouraging environment.

Math Tutor *January 2025-present*

Projects

Hospital Management System

2022 Oxford Secondary School

Hackathon

As team lead, I developed a centralized platform to manage hospital workflows, patient records, and administrative scheduling. The system was designed to streamline medical operations, with proposed future enhancements including e-prescriptions and automated doctor-matching for faster patient care. The project earned fourth place out of 19 competing teams.

Dual Authentication System
2025

Wright Hackathon

As part of a 24-hour hackathon challenge, I collaborated with a team to build a secure dual authentication system integrating a mobile app and web interface to protect user data. The system implemented access-control features ensuring identity verification before accessing critical data, demonstrating rapid prototyping, innovative problem-solving, and strong teamwork. The project was awarded Best Collaboration Team at the Make It Wright Hackathon 2025.

Technical Skills

Languages: Java, Python, C, SQL, JavaScript, PHP, Shell, Assembly, QBASIC

Web Development: HTML, CSS, React, Tailwind CSS

Tools: Git, GitHub, VS Code, Linux, MS Office, IntelliJ, Replit, DevC++, CLion

Soft Skills: Tutoring, Collaboration, Communication, Problem Solving, Leadership

Awards and Recognition

CRLA Level I Certified Tutor, Wright State University MLC — 2025

Technology First 2025 Scholar

Recognized for Creativity & Collaboration — Make It Wright Hackathon 2025

Fourth Place Hackathon Ranking (out of 19 teams) 2022 Oxford Secondary School Hackathon

Leadership & Community Engagement

Volunteer - Blood Donation Programs (Red Cross & Lions Club); led blood donation camp organization during high school.

School Quiz Team Representative, Grades 7-10 - competed in regional inter-school quiz competitions.

Science Exhibition Participant - created a prototype device to convert air pollution into usable acidic compounds for cosmetic and industrial uses.

Daniel Evans

Dayton, OH | (937) 896-2276 | evansd161@gmail.com | *U.S. Citizen*

Current Mechanical Engineering student at Wright State University ready to apply problem-solving skills and technical skills. Looking to contribute to engineering projects while gaining valuable experience in the field.

Education

Wright State University, Dayton, OH Aug. 2024–Present

Bachelor of Science in Mechanical Engineering

Mechanical engineering coursework with a focus on high-level math, physics, and engineering analysis. Relevant coursework includes SolidWorks, Matlab, Statics, and Properties of Materials.
GPA: 3.934

Edison State Community College, Piqua, OH Aug. 2022–May 2024

STEM Coursework

Completed science and engineering coursework that built on my experience at Graham High School taking Honors and CCP courses.
GPA: 3.90

Projects

Matlab Application Design May 2025

Designed and developed four interactive applications using Matlab App Designer. Collaborated with a partner to design, test, and troubleshoot code, ensuring functionality across all applications.

Cardboard Bridge Design Dec. 2024

Designed, built, and tested a cardboard bridge with a partner, using SolidWorks for simulation and evaluating its strength and load capacity.

3D-Printing July 2024

Took part in a 3D printing project, gaining an understanding of the design process. Designed and created 3D-printed models using SolidWorks while learning design optimization.

Technical Skills

SolidWorks (Certified SolidWorks Associate – Dec. 2024)

LTspice, Matlab, R

Awards and Recognition

Recipient of Mercy Health's Give for Good Scholarship

Recipient of the 4-H Family and Consumer Sciences Scholarship

Wright State University Dean's List (3 semesters)

Edison State Community College Dean's List (1 semester)

Erica Gilliam

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I am a systems-focused professional with a work history that illustrates my commitment to learning and growth. I am a fast learner who enjoys and thrives in fast-paced environments. I'm dedicated to continued improvement, personally and professionally.

Education

Wright State University, Dayton, OH

Aug. 2025–present

Bachelor of Science in Industrial and Systems Engineering

Coursework for this degree, which is focused on improving the efficiency and quality of systems, is centered on high-level math, physics, and statistics. Classes taken to date include optimization, human fundamentals, advanced statistics, Six Sigma, Solid Works, Python, Matlab, and R.

GPA 3.7

Sinclair College, Dayton, OH

Aug. 2022–Aug. 2025

STEM Coursework

Classes taken in science and engineering built on my experience at Kettering High School taking Honors and AP courses in their engineering program.

Experience

The TSP Solution, LLC, Dayton, OH

TSP Solution is a contracted delivery service provider (DSP) exclusive to Amazon at their Kettering, Ohio warehouse. In my current role, I manage a fleet of vans to support assigned routes, which vary daily and widely (from 35–75); monitor and adjust routes and/or drivers throughout the day to maintain Amazon's targeted performance; track vehicle maintenance and file all affiliated reporting; and investigate damage and other incident reports and work with the reporting parties toward a positive outcome.

Fleet Manager

April 2025–present

Team Lead

Oct. 2024–April 2025

Delivery Driver

April 2021–Oct. 2024

The Kroger Co., Dayton OH

This Cincinnati-based grocery retailer operates more than 2,700 stores through the U.S., many of which include pharmacies and additional retail. One of the largest retailers in the country, they are a leader in developing best practices in inventory and in-store technologies.

Assistant Drug/General Merchandise Manager

May 2020–April 2021

Grocery Clerk

March 2019–May 2020

Courtesy Clerk

June 2017–March 2019

Technical Skills

Six Sigma Green Belt, Solid Work, Python, Matlab, R

Awards and Recognition

Recipient of the Wright State University Rike Associate Degree Transfer Scholarship, which recognizes academic performance

Sinclair College Dean's List for 6 of 8 quarters

Reese Harmeyer

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I am a computer engineering student that is currently studying at Wright State. I am very good at adapting to different environments and can also learn complex processes very quickly. I have good work ethic, and I do things to the best of my ability.

Education

Wright State University, Dayton, OH

Aug. 2025–present

Bachelor of Science Computer Engineering

Coursework for this degree, which is focused on understanding digital and electrical systems as well as digital logic. It is centered on high-level math, computer science, and statistics. Classes taken to date include devices and circuits, digital circuit design, computer organization, and technical communications. I currently have a 4.0 GPA.

Sinclair College, Dayton, OH

Aug. 2022–May. 2024

Engineering Coursework

I took gen ed courses like psychology and anthropology as well as the basic math and science courses that engineers need a framework to build from. I took AP Calculus at Carlisle High School which allowed me to take some extra courses.

Experience

Camp Chautauqua Inc., Miamisburg, OH

I currently work as a maintenance assistant. My main responsibility is to handle any sort of electrical project that the camp is doing whether we are doing new construction, putting up new lights, ballast bypassing, or repairing a new issue. When there isn't electric work to be done, I do various other tasks such as: maintaining equipment, framing, and plumbing.

Maintenance Assistant

August 2024–present

Intern

May 2022–June 2022

Grill Manager

May 2021–August 2021

Kings Electric., Lebanon OH

I worked at Kings Electric for about 4 months as a sort of summer internship. I learned electrical code, how to bend conduit, and how to use a hammer drill to work with concrete.

Apprentice Electrician

May 2023–August 2023

Technical Skills

C++, Java, Matlab, VHDL, Rockwell Software

Awards and Recognition

Associates Transfer Scholarship

Carter Rockenbaugh

| Dayton OH | (937) 578-3383 | carterrockenbaugh@gmail.com | U.S. Citizen |

I am a systems-focused professional with a work history that illustrates my commitment to learning and growth. I am a fast learner who enjoys and thrives in fast-paced environments. I'm dedicated to continued improvement, personally and professionally.

Education

Wright State University, Dayton, OH Aug. 2025–present

Bachelor of Science Mechanical Engineering

Coursework for this degree, which is focused on the design, analysis, and improvement of mechanical systems, is centered on high-level math, physics, and engineering principles. Classes taken to date include statics, dynamics, thermodynamics, fluid mechanics, materials science, SolidWorks, Python, and MATLAB.

Experience

Walmart Supercenter, Urbana OH

Walmart is a multinational retail corporation offering grocery and general merchandise services, including Online Grocery Pickup (OGP). In this role, I supported daily OGP operations by picking and staging customer orders, ensuring accuracy and quality; assisted with dispensing orders to customers; and collaborated with team members to maintain efficient workflow during high-volume periods.

Associate

November 2022–May 2024

Hilton Garden Inn, Beavercreek OH

Hilton Garden Inn is a full-service hotel brand focused on guest comfort and reliability. In this role, I performed routine maintenance and repairs across hotel facilities, troubleshooted mechanical and electrical issues, and responded to guest service requests to ensure a safe, functional, and comfortable environment.

Maintenance Technician January 2025 – July 2025

Hotel Ardent by Hilton

Hotel Ardent is a modern hospitality brand focused on delivering high-quality guest experience. In this role, I performed routine maintenance and repairs, troubleshooted mechanical and electrical issues, and responded to service requests to ensure a safe, functional, and well-maintained environment.

Engineer August 2025 – Present

Technical Skills

Solid Work, MATLAB

Awards and Recognition

AFROTC Scholarship Recipient; Recognizes a mix of academic ability, leadership potential, and overall character

Facilities and Equipment

We will be partnering with Wright State University. They will provide an adequate space for our research and development of a prototype cleaning robot.

We will need the following resources:

- 2 SolidWorks Licenses
- 5 Desktop computers capable of running SolidWorks and other intense programs
- 1 Arduino
- 1 Electric motor
- 2 Radial brushes
- 1 Water pump
- 2 Water jets
- 1 Battery
- Miscellaneous supplies for the cleaning robot prototype

Budget and Justification

Our proposed budget is Two hundred fifty thousand dollars; this would be for research of the floating solar farms and the prototype cleaning robot. A large portion of this budget would be allocated to the development of the cleaning robot, this would include research, design in 3d modeling, and prototype fabrication of the robot. These tests would ensure the system can reliably operate in harsh weather and marine conditions, the success of this robot in the testing and development phase are paramount. Specifically, the cleaning robot will need the following: An Arduino, an electric motor, radial brushes, a water pump, water jets, and a battery. The Arduino will cost \$22.80 (Audino, n.d.)

Additionally, we would need to use funds to pay the wages of hired individuals. The projected positions needed would be: 1 computer scientist, 1 computer engineer, 1 industrial engineer, and 2 mechanical engineers, who would be responsible for the design, construction and testing of the equipment and systems. They will each be paid \$20 per hour. A smaller portion of the fund would be used to acquire the unknown materials that will be needed for the cleaning robot. We will also have to pay for the design tools we use; such as SolidWorks which costs \$2320 per license (Dassault Systèmes, 2025.) Computers that can run intense software will also be required and will cost \$2999 per computer (HP Development Company, 2026.)

The remainder of our budget will be an emergency fund in case of design modifications or unforeseen occurrences that would hinder our progress. Overall, this budget would allow ample funds to ensure the success of our project, from initial to final testing, and support flexibility for unforeseen challenges.

Category	Item	Number	Subtotal	Total
Personnel	Computer Scientist Salary, \$20/hour	2080 hours	\$41600	\$41600.00
Personnel	Computer Engineer Salary, \$20/hour	2080 hours	\$41600	\$83200.00
Personnel	Industrial Engineer Salary, \$20/hour	2080 hours	\$41600	\$124800.00
Personnel	2 Mechanical Engineers Salary, \$20/hour	4160 hours	\$83200	\$20800.00
Equipment	SolidWorks (Dassault Systèmes, 2025)	2	\$5640	\$213640.00
Equipment	Desktop Computers (HP Development Company, 2026)	5	\$14995	\$228635.00
Equipment	Arduino (Arduino, n.d.)	1	\$22.80	\$228657.80
Supplies	Electric Motor	1	\$500	\$229157.80
Supplies	Radial brushes	2	\$500	\$229657.80

Supplies	Water Pump	1	\$100	\$229757.80
Supplies	Water Jets	2	\$75	\$229832.80
Supplies	Battery	1	\$300	\$230132.80
Supplies	Miscellaneous Cleaning Robot Materials	N/a	\$15000	\$245132.80

References

- Arduino. (n.d.). *Arduino Micro*. Retrieved April 23, 2026, from <https://store-usa.arduino.cc/products/arduino-micro>
- Dassault Systèmes. (2025, May). *SOLIDWORKS design plans & pricing*. <https://www.solidworks.com/how-to-buy/solidworks-design-plans-pricing>
- HP Development Company. (2026). *HP Z2 Tower G1i desktop workstation with 3 yr warranty & Wolf Pro Security*. <https://www.hp.com/us-en/shop/pdp/hp-z2-tower-g1i-workstation-desktop-pc-wolf-pro-security-edition-p-bp8m3ua-aba-1>
- Huang, L., Elzaabalawy, H., Sarhaan, M., Sherif, A., Ding, H., Ou, B., Yang, D., & Cerik, B. C. (2025). Developing reliable floating solar systems on seas: A review. *Ocean Engineering*, 322, Article 120525. <https://doi.org/10.1016/j.oceaneng.2025.120525>
- Kekre, A., & Gawre, S. K. (2017). Solar photovoltaic remote monitoring system using IoT. *International Conference on Recent Innovations in Signal Processing and Embedded Systems (RISE)*, Bhopal, India, 508–512. <https://doi.org/10.1109/RISE.2017.8378227>
- Mani, M., & Pillai, R. (2010). Impact of dust on solar photovoltaic (PV) performance: Research status, challenges and recommendations. *Renewable and Sustainable Energy Reviews*, 14(9), 3124–3131. <https://doi.org/10.1016/j.rser.2010.07.065>
- Nedelcu, A.-T., Faităr, C., Voicu, I., & Panaitescu, M. (2026). Floating photovoltaic systems—Energy performance and environmental challenges in sustainable development. *Sustainability*, 18(5), Article 2588. <https://doi.org/10.3390/su18052588>
- OpenAI. (2026). *Gantt chart of solar panel cleaning robot project timeline* [AI-generated image]. ChatGPT. <https://chatgpt.com/>
- Thangarasu, S. K., Gobinath, V. K., Shanmugam, A., Kumar, S. M., Shrinidhi, K. M., & Sanjay,

S. (2024). Innovative Arduino-based solar panel cleaning system for enhanced energy efficiency. *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, Kamand, India, 1–5.

<https://doi.org/10.1109/ICCCNT61001.2024.10725904>

Xiong, Y., Liu, H., Xu, J., & Chen, H. (2025). Mechanical properties of offshore floating photovoltaic structural coating/carbon steel system in a marine environment. *Ocean Engineering*, 323, Article 120585. <https://doi.org/10.1016/j.oceaneng.2025.120585>

Zeng, F., Bi, C., Sree, D., Huang, G., Zhang, N., & Law, A. W.-K. (2023). An adaptive barrier mooring system for coastal floating solar farms. *Applied Energy*, 348, Article 121618. <https://doi.org/10.1016/j.apenergy.2023.121618>